Topics covered on ML and DL By Andrew Ng

- Linear Regression with One Variable

It means that the regression model is concerned with two dimensions only, one for input and another for output.

Basically, the x value is taken as input for the model and y value gives us the output.

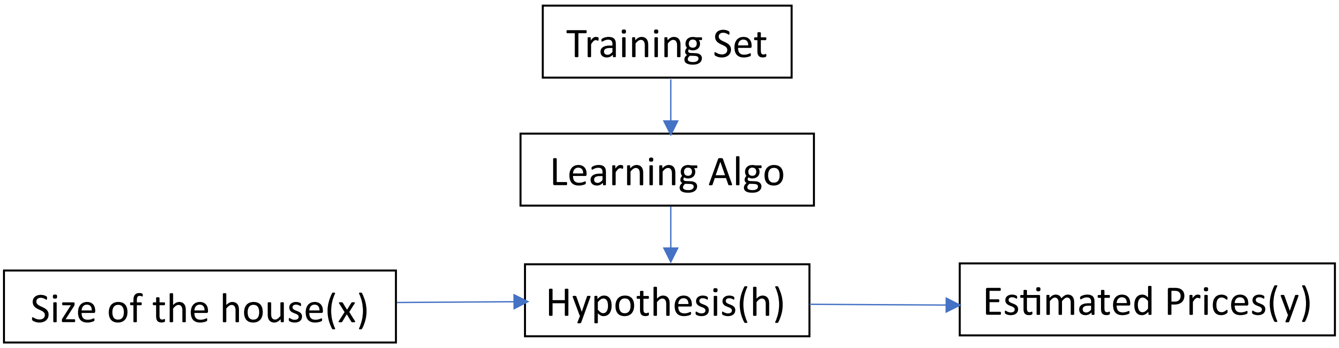
- Supervised problem

Give the right answer for each example in the data

- Regression Problem

Predict real-valued output

Example of House price Prediction



Optimizers

As the name suggest, optimizers are used for optimizing our model.

They are used to adjust the parameters in the model so that the errors and loss functions decrease and we get a better result than the previous one.

The Optimizers are:

1) Gradient Descent
2) Stochastic Gradient Descent (SGD)
3) Adam
4) RMSprop

- Gradient Descent

When you have some function $J(\theta_1, \theta_2)$ and you want the minium value of $J(\theta_1, \theta_2)$, we use gradient descent.

- Batch Gradient Descent

"Batch": each step of the gradient descent uses all the training examples .

- Feature Scaling

Make sure features are on a similar scale.

- Mean Normalization

Replace x_i to $x_i - \mu_i$ to make features have approximately 0 mean .

- Normal Equation

Method to solve for Θ analytically.

- Difference Between Gradient Descent and Normal Equation

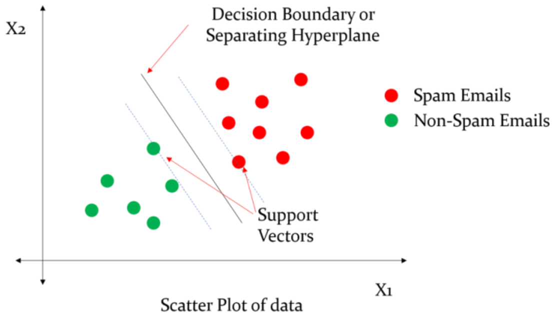
Gradient Descent	Normal Equation
Need to choose α.	No need to choose α.
Need many iteration.	Don't need to iterate.
Works well even when n is large.	Works slow when n is large.
$n = 10^6$	n = 100 n = 1000 n = 10000

- Stochastic Gradient Descent
  Much faster than Gradient Descent.
  Instead of calculating the gradient for the whole dataset, it calculates the gradient for small randomly selected dataset batches.
  It leads to randomness, efficiency, and helps in reducing noise.
- Adam Optimizer
  A popular optimizer in machine learning models.
  This optimizer takes actions according to the magnitude of the recent gradients which ultimately helps in shallowing the local minima.
- RMSprop
  Stands for Root Mean Square Propagation
  Used to overcome the shortcomings of the traditional Gradient Descent.
  Formula -> $E[g^2] = p * E[g^2] + (1 – p)* g^2$
  g = gradient of current parameter
  $E[g^2]$ = accumulated square gradient.
- Decision Boundary

The line which separates one class from another class.

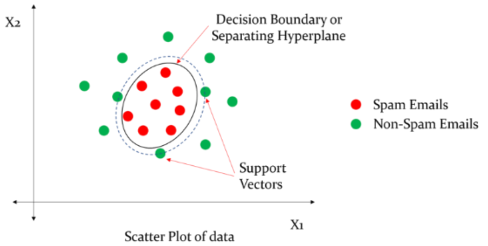
Can be classified into two types

1) Linear Decision boundary
   a) Used in linear models such as linear regression and support vector machine .
   b) These create straight line boundaries.



Linear Decision Boundary

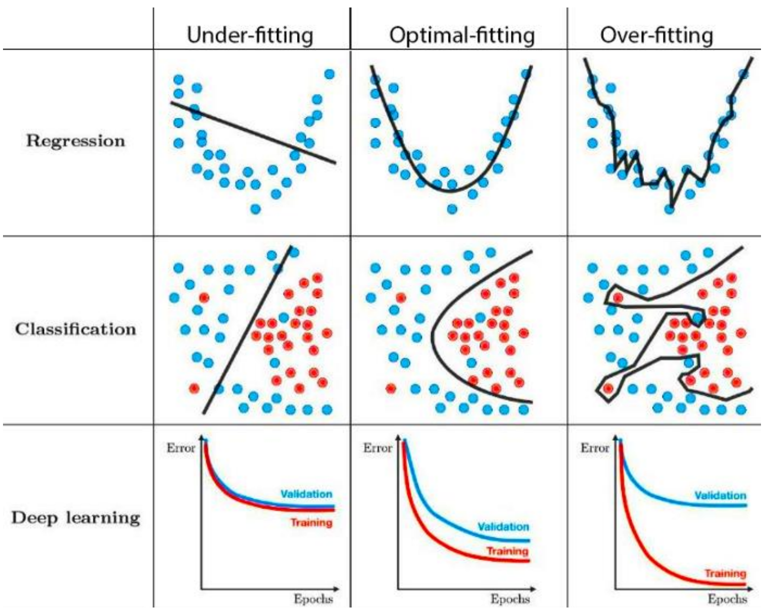
2) Non-Linear decision boundary
   a) Used in non-linear models such as random forest, decision trees, neural networks etc.
   b) These create more complex and curved boundaries.



Non- Linear Decision Boundary

- Regularization

This process is basically used to solve the problem of over-fitting while mapping linear regression, classification and deep learning classes.



- What is Over-fitting?

When we have too many features, the learned hypothesis may fit the training set very well, but fail to generalize to new examples (predict precisely on new examples).

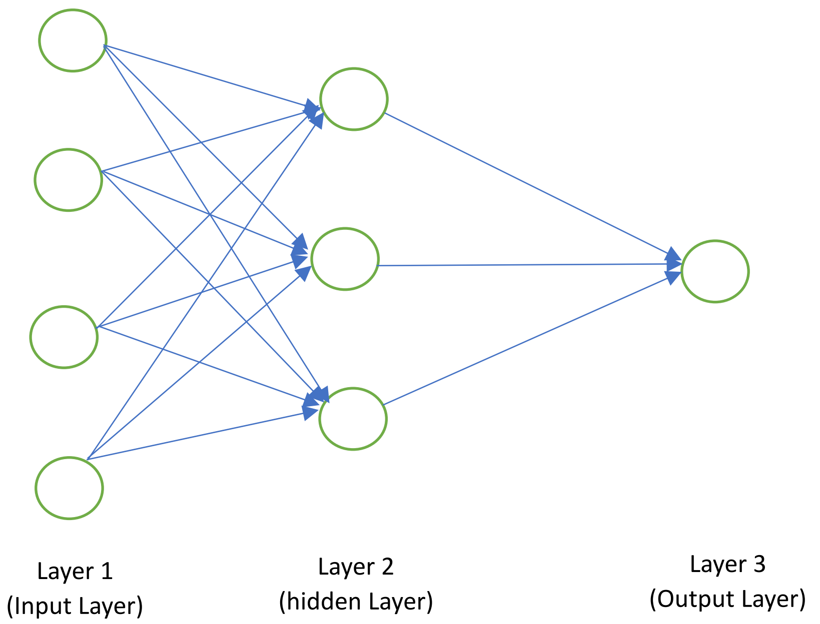
This situation basically occurs when we try to fit a large amount of features to training and testing models.

Options to overcome overfit:-

1) Reduce number of features
   - Manually select which features to keep.
   - Use Model selection algorithm.
2) Regularization
   - Keep all the features, but reduce magnitude / values of parameters.
   - Works well when we have a lot of features, each of which contributes a bit to predicting y.

- Neural Network Representation

Origin: Algorithms that try to mimic the brain.



- Input Layer

We give out input values and parameters to the model in this layer .

- Hidden Layer

The calculations inside the model take place in this layer.

- Output layer

The result is given out from the model through this layer.

- Weights and Biases

Weights determine the importance of each input.

Biases are additional parameters that help adjust the output along with weighted sum of inputs.

- Cost Function(J(Θ))

It is a measure of how well a model's prediction match the actual data. It quantifies (to express or measure) the error between the predicted values and the actual values, helping to evaluate the performance of the model.

- The goal is to minimize the cost function to improve the model's accuracy.

In linear regression the cost function(J(Θ)) helps in finding the best fit line by minimizing the sum of the squared differences between the predicted and actual value.

Problem:

While testing your hypothesis on a new set of houses , you find that it can make unacceptably large errors in its predictions.

Solution:

a) Get more training examples.
b) Try smaller set of features.
c) Try getting additional features
d) Try adding polynomial features (x_1^2, x_2^2, $x_{1,}$ x_2 etc.)
e) Try decreasing λ.
f) Try increasing λ.

- Precision and Recall

True Positive	False Positive
False Negative	True Negative

Let's take an example of cancer prediction model !!

Precision

Of all the patients where we predicted y = 1, what fraction actually has cancer ?

True positives / predicted positives = True Positives / True pos + False pos

Recall

What fraction did we correctly detect as having cancer ?

True positives / actual positives = True positives / True pos + False neg

F1 score

Used to compare different algos based on the precision and recall

F1 score = 2PR/P+R

- Support Vector Mechanism (SVM)

Support Vector Machines (SVM) are a set of supervised learning methods used for classification, regression, and outliers detection.

It works by separating the classes with the help of a hyperplane in a high dimentional plane.

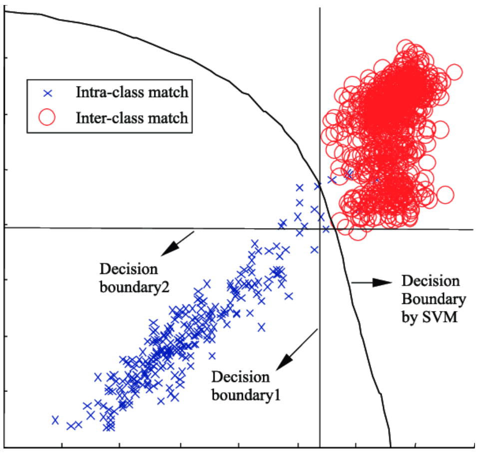
SVM hypothesis

Min $\Sigma[y^{(i)} \text{cost}_1(\Theta^T x^{(i)}) + (1-y^{(i)})\text{cost}(\Theta^T x^{(i)})] + \frac{1}{2}\Sigma\, \Theta^2 j$

If y = 1 , we want $\Theta^T x \geq 1$ (not just ≥ 0) $\Theta^T \geq \emptyset^1$

If y = 1 , we want $\Theta^T x \leq 1$ (not just < 0) $\Theta^T \leq \emptyset^1$

SVM decision boundary

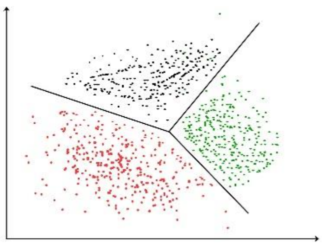


Kernels

Given a feature x, they compute new features depending on the proximity to the landmarks on the graph.

Clustering [Unsupervised Learning]

A type of unsupervised learning techniques which is used to find groups using given points in a graph .



Application of clustering:-

1) Market segmentation
2) Social Network analysis
3) Organize computing clusters
4) Astronomical data analysis

- K-means Algorithm

Input:

- ➔ K(number of custers)
- ➔ Training set $\{x^{(1)}, x^{(2)}, x^{(3)}, ..., x^{(m)}\}$
- ➔ $x^{(i)} = R^n$ (drop $x_o = 1$ convention)

Algorithm:

Randomly initialize K clusters centroid $\mu_{1,} \mu_{2,} \mu_{3} ..., \mu_{k} = R^n$

Repeat {

Cluster assignment step

For i=1 to n

$C^{(i)}$:= index(from 1 to k) of cluster centroid closest to $x^{(i)}$

Move Centroid

For k = 1 to k

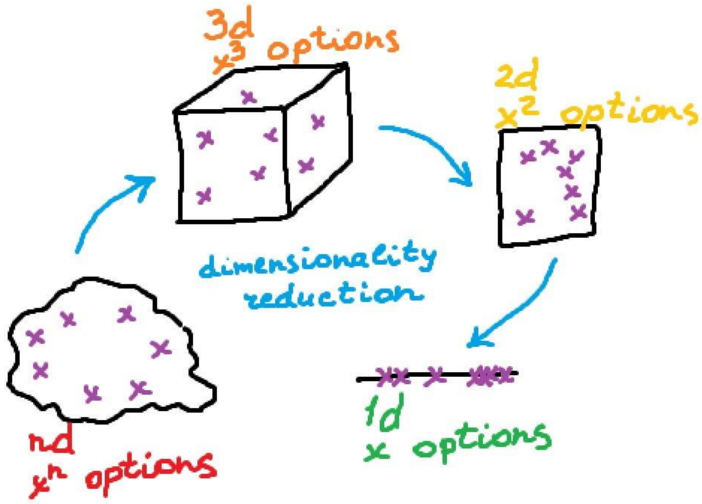
μ_k := average (mean) of points assigned to cluster k

}

Dimentionality Reduction

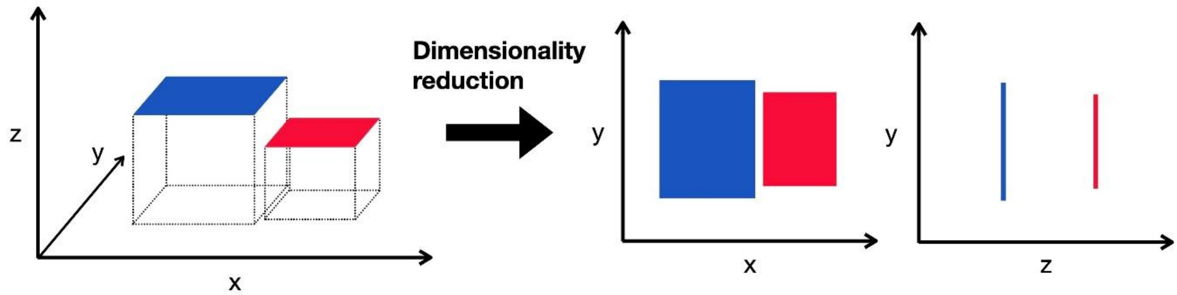
Motivation 1: Data compression

Reduce data from 2D to 1D or from 3D to 2D.



Motivation 2: Data Visualization

Suppose we have a dataset of a country's GDP growth , Human Development , Life expectancy , poverty index etc. By using the dimentionality reduction process , we can easily classify the above parameters from to two parameters (such as Z_1 and Z_2) .



PCA (Principal Component Analysis)

It is a type of dimentionality reduction technique . The PCA finds the principal components where data varies the most.

Choosing K (The member of principal components)

Average squared projection error : $(1/m).\Sigma||x^{(i)} - X^{(i)}_{approx.}||$ -- (i)

Total variation in the data: $(1/m)\ \Sigma||x^{(i)}||^2$ -- (ii)

Typically, choose k to be the smallest value so that :

$$(i)/(ii) \leq 0.01$$

"here 99% of the variance is retained"

Application of PCA

- ➔ Compression
  - Reduce memory / disk needed to store data.
  - Speed up learning algo.

  (Remember: choose k by % of variance retain)

- ➔ Visualization

Bad use of PCA

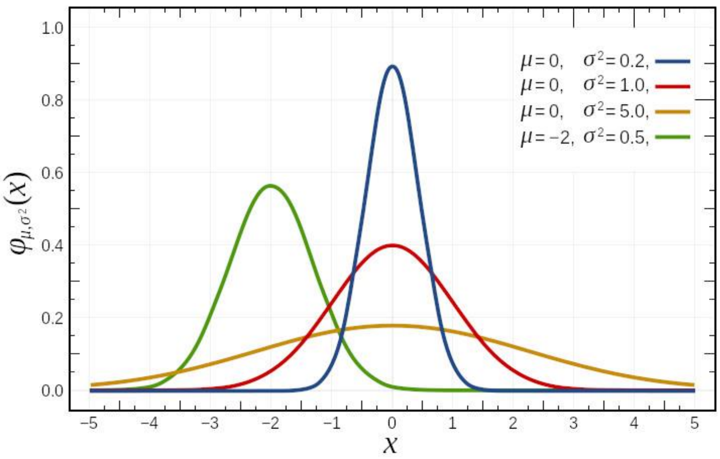
- ➔ Fewer features less likely to overfit.

Gausian Distribution

The gaussian distribution is mainly defined by the bell shaped curve which is normally constructed using two parameters namely mean and standard deviation . It is also known as normal distribution curve.

Mean(σ): it is a central value in the graph where peak of the curve is located.

Standard deviation(μ): measures the spread of the curve



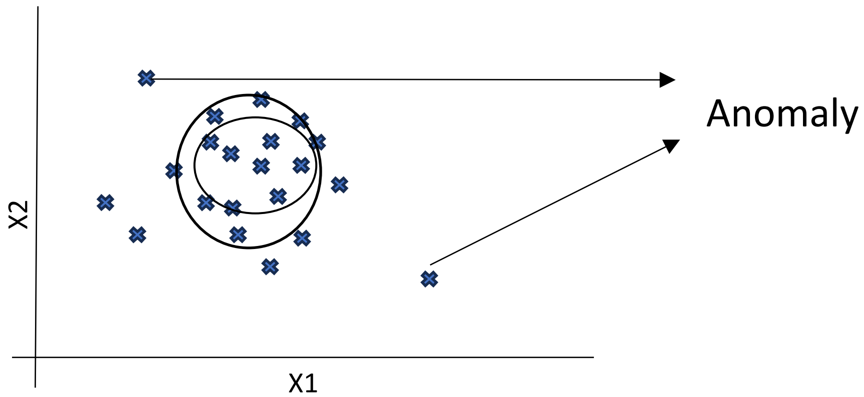
Anomaly Detection

It is a process of identifying unusual or unexpected patters in the dataset when mapped. The common applications of anomaly detection are fraud detection, network security, manufacturing fault detection etc.

Illustration:-

X1 = heat generated

X2 = vibration intensity



Anomaly detection algorithm

1) Choose features x_i that you think might be indicative of anomalous examples.
2) Fit parameters μ_1 , μ_2 ……, μ_n, σ_1^2, ….. , σ_n^2

   $\mu_j = 1/m\ \Sigma\ x_j$

$\sigma_j^2 = 1/m\ \Sigma\ (x_j^{(i)} - \mu_j)^2$

3) Given new example compute p(x)

   $P(x) = \Pi\ p(x_j, \mu_j, \sigma_j^2)$

   Anomaly if $p(x) < \varepsilon$